2025 Physics Final Exam

About the document:

- This exam is from the 2025 Final Exam (25 Nov 2025), Section 2 (NANO), and it is not the full exam. I have <u>only included notable problems</u> (No question here is from the in-class exercise set).
- I memorized all these questions personally inside the exam room, please accept that there may be minor errors.
- Although these questions are purely written from memory, I have retained the exact numbers and values used in the actual exam, so treat the math as exact.

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Constants and conversions

$g = 9.80\ \text{m/s}^2$

$N_A = 6.02\times10^{23}$ molecules/mol

$k_B = 1.38\times10^{-23}$ J/K

$R = 8.314$ J/(mol•K)

1 cal = 4.186 J

1 bar = 10^5 Pa

Trigonometric formulae

$$\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$$

$$\cos A + \cos B = 2\cos(\frac{A+B}{2})\cos(\frac{A-B}{2})$$

Fluid mechanics

Hydrostatic pressure: $P = P_0 + \rho g h$

Continuity equation: $A_1 v_1 = A_2 v_2$

Bernoulli's equation for an ideal fluid:

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{const}$$

Buoyant force: $F_B = \rho_{\text{fluid}} g V_{\text{disp}}$

Simple harmonic motion

$$f = \frac{1}{T},\ \omega = \frac{2\pi}{T} = 2\pi f,\ U_{\text{gravity}} = mgh,$$

$$U_{\text{spring}} = \frac{1}{2}kx^2,\ K_{\text{translation}} = \frac{1}{2}mv^2,\ K_{\text{rotation}} = \frac{1}{2}I\omega^2$$

$$I = I_{CM} + M d^2,\ E_{\text{SHM}} = U + K = \text{const}$$

Natural frequency of oscillations:

$$\omega = \sqrt{\frac{k}{m}};\ \omega = \sqrt{\frac{g}{L}};\ \omega = \sqrt{\frac{mgd}{I}};\ \omega = \sqrt{\frac{\kappa}{I}}$$

Damped oscillation: Retarding force $\vec{R} = -b\vec{v}$

$$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0$$

$$x(t) = Ae^{-\frac{b}{2m}t}\cos(\omega t + \phi);$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2};\ \omega_0 = \sqrt{\frac{k}{m}}$$

Forced oscillation:

$$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = \frac{F_0}{m}\sin(\omega t)$$

$$x(t) = A\cos(\omega t + \phi);$$

$$A = \frac{F_0/m}{\sqrt{\left(\omega^2 - \omega_0^2\right)^2 + \left(\frac{b\omega}{m}\right)^2}}$$

Waves and Sound

Traveling wave: $y(x,t) = A\sin(kx \mp \omega t + \phi_0)$

Rate of energy transfer of mechanical waves:

$$P = \frac{1}{2}\mu\omega^2 A^2 v$$

Standing waves (nodes at both ends, string length L):

$f_n = \frac{nv}{2L}$; $n =$ 1 (fundamental); 2, 3, ... (harmonics)

Air column opened at one end and closed at the other: $f_n = \frac{nv}{4L}$; $n =$ 1, 3, 5, ... (odd harmonics)

Sound level (dB) $\beta = 10\log\left(\frac{I}{I_0}\right)$, $I_0 = 10^{-12}$ W/m^2

Doppler effect: $f' = f\left(\frac{v \pm v_o}{v \mp v_s}\right)$

Supersonic:

Mach number: $M = \frac{v_s}{v}$; Mach cone: $\sin\theta = \frac{1}{M}$

Beat: $f_{\text{beat}} = |f_1 - f_2|$

Heat and Thermodynamics

Absolute $T = T_{^\circ C} + 273.15$

Linear thermal expansion: $\Delta L = \alpha L_0 \Delta T$

Heat conduction: $P = \frac{A(T_H - T_C)}{\sum_i R_i}$, $R_i = \frac{L_i}{k_i}$

Ideal gas law: $PV = Nk_BT = nRT$

Adiabatic process: $PV^\gamma =$ const, $\gamma = \frac{c_P}{c_V} = \frac{c_V + R}{c_V}$

1st law of thermodynamics:

$$\Delta E_{\text{int}} = Q + W,\ dE_{\text{int}} = dQ + dW$$

Heat: $Q = mc\Delta T$; $Q = mL$; $Q = nc_P\Delta T$; $Q = nc_V\Delta T$

Work: $W = -\int PdV$

$$W = nRT\ln\left(\frac{V_i}{V_f}\right) = nRT\ln\left(\frac{P_f}{P_i}\right)$$

$$W = \frac{1}{\gamma - 1}(P_fV_f - P_iV_i)$$

Maxwell-Boltzmann distribution:

$$N_v = 4\pi N\left(\frac{m}{2\pi k_BT}\right)^{\frac{3}{2}} v^2 e^{\frac{-mv^2}{2k_BT}}$$

$$v_{mp} = \sqrt{\frac{2k_BT}{m}} = \sqrt{\frac{2RT}{M}},\ v_{avg} = \sqrt{\frac{8k_BT}{\pi m}} = \sqrt{\frac{8RT}{\pi M}}$$

$$v_{rms} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3RT}{M}}$$

Thermal efficiency: $e \equiv \frac{W_{\text{eng}}}{|Q_{\text{h}}|} = \frac{|Q_{\text{h}}| - |Q_{\text{c}}|}{|Q_{\text{h}}|}$

Heat pump COP (cooling) $= \frac{|Q_{\text{c}}|}{W}$

Heat pump COP (heating) $= \frac{|Q_{\text{h}}|}{W}$

Carnot engine : $e = 1 - \frac{T_c}{T_h}$, $\frac{T_c}{T_h} = \frac{|Q_c|}{|Q_h|}$

Change of entropy:

$$\Delta s = \int\frac{dQ}{T},\ \Delta s = nR\ln\left(\frac{V_f}{V_i}\right) + nc_V\ln\left(\frac{T_f}{T_i}\right)$$

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Consider a cubic box of mass M and side length D that is fully submerged in a non-uniform density liquid. The top face of the box is level with the surface of the liquid. The density within the liquid follows the distribution of

$$\rho(y) = \rho_0 + \rho_1 y$$

where y is the depth measured from the surface, and ρ_0 and ρ_1 are constants.

An external vertical force of magnitude F is applied to the top of the box to maintain it in equilibrium.

A. Determine the hydrostatic pressure at the bottom of the box.
B. Determine the magnitude of the external force F required to keep the box in equilibrium.

Answers in terms of D, g, ρ_0, ρ_1, M

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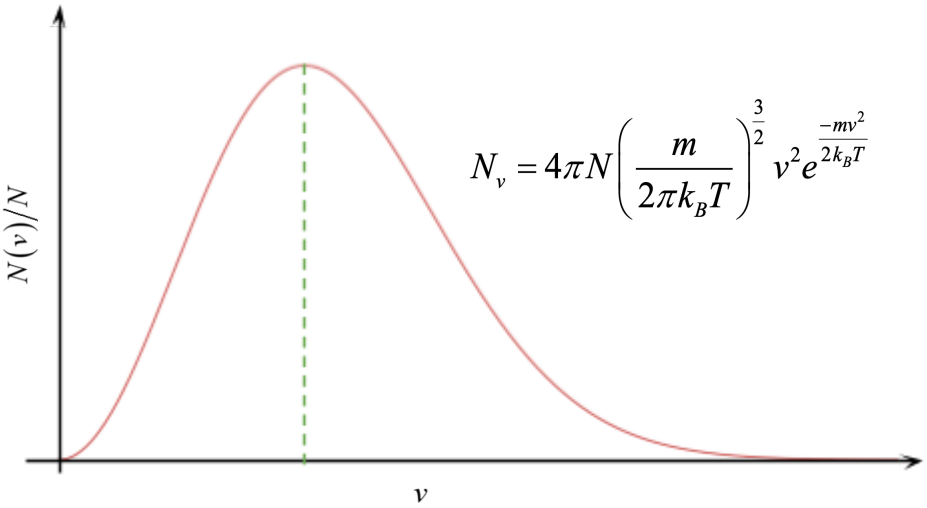
The figure below illustrates the Maxwell-Boltzmann speed distribution for oxygen molecules, where $N(v)dv$ represents the number of particles with speeds in the interval $[v,\ v\ +\ dv]$, and N is the total number of particles. The vertical axis represents $N(v)/N$, the horizontal being v.



Assuming the system is in thermodynamic equilibrium at $300K$

A. Calculate the most probable speed of the molecules in the system.
B. Determine the peak particle density, N, at this speed in A.
C. Sketch the new normalized distribution curve, $N(v)$/N, if the temperature is increased significantly above 300K, and explain the changes.

Suppose the molecular weight of an oxygen molecule (O_2) is $31.998\ [g/mol]$

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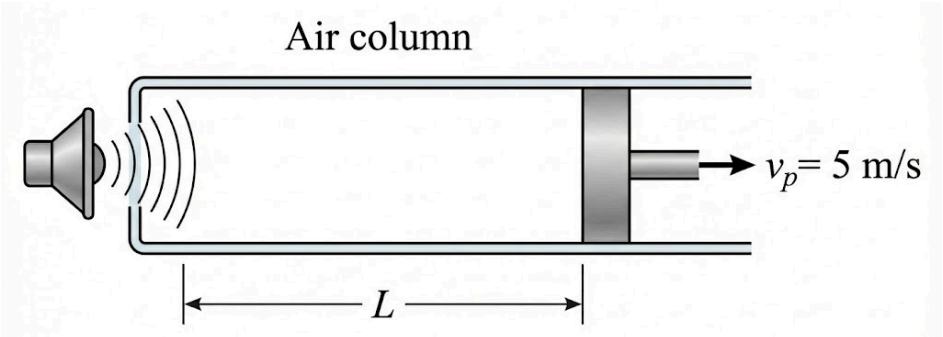
A sound source emitting a frequency of $1200\ Hz$ is positioned at the open end of a tub. The other end is sealed by a movable piston. At $t = 0$, the piston beings moving away from the source at a constant velocity of $5\ m/s$. The speed of sound in air is given by $v = 331 + 0.61T\ [m/s]$, where T is the temperature in degrees Celsius.



A. Assuming the air temperature is $25\ °C$, determine the length of the air column required to support the fundamental resonance mode.
B. Calculate the time interval between two consecutive resonance events at $25\ °C$
C. If the system is heated to $80\ °C$, determine whether this time interval increases or decreases compared to the interval at $25\ °C$. Calculate the magnitude of this difference.

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A simple pendulum consists of a spherical bob of radius r and density of ρ_m, attached to a massless string of length L. the entire system is fully submerged in a non-ciscous liquid of density ρ (where $\rho < \rho_m$)

A. Derive the general expression for the angular frequency ω of small oscillations for this system ,accounting for the finite size of the sphere.
B. From part A: Assuming the string length is much greater than the bob's radius ($L >> r$), determine the simplified expression for the angular frequency ω.

Assume the drag force from the liquid is negligible.
(You are not required to solve differential equation explicitly)

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Three individuals, A, B, and C, are positioned in a straight line such that B is equidistant from A and C. They hold tuning forks with frequencies of $503\ Hz,\ 505\ Hz,\ 507\ Hz$, respectively. Assume the speed of sound is $343\ m/s$

A. If A and C strike their tuning forks while all individuals are stationary, determine the average frequency heard by B.
B. If A strikes his tuning fork and B run towards A at $5\ m/s$, calculate the frequency observed by B.
C. If both A and C strike their tuning forks, determine the speed and direction B must run in order to hear a single continuous tone (zero beat frequency).

(Note: Constants value in this question are remade, as I can't recall them)

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